

Functional and differential equations

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Second-order differential inequalities and estimates of solutions

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Theoretical minimum.

Theorem 1 (explicit bounds from an inequality). If L is a linear differential operator with constant coefficients whose characteristic roots are real, then $Lf \geq 0$ together with initial data forces f to dominate the solution of $Ly = 0$ with the same data: one factors $L = (\frac{d}{dx} - \lambda_1)(\frac{d}{dx} - \lambda_2)$ and integrates the two first-order inequalities in turn. This is how $f'' - 5f' + 6f \geq 0$ produces an explicit lower bound for f .

Theorem 2 (Riccati and the length of the interval). The substitution $u = \frac{y'}{y}$ turns $y'' + qy = 0$ into $u' + u^2 + q = 0$ (Differentiable functions – 6), and conversely an inequality $u' + u^2 \geq -q$ on an interval where u runs from $+\infty$ to $-\infty$ forces the interval to be long: for $q \equiv 1$ its length is at least π , because $u = \cot$ is the extremal case.

Theorem 3 (the maximum principle). If $y'' \geq 0$ on $[a, b]$, then y attains its maximum at an endpoint; if $y'' + qy \geq 0$ with $q \leq 0$, the same holds. Consequently a solution of $y'' = qy$ with $q \geq 0$ vanishing at both endpoints is identically zero — the uniqueness statement for the boundary value problem, obtained without any formula.

Fact 1 (three sources of estimates). Gronwall's lemma for first-order problems (Differentiable functions – 6); comparison with an explicit solution for second-order ones; and integral inequalities of Wirtinger, Poincaré and Opial type when the quantity to be bounded is an integral of the solution (Differentiable functions – 9). Choosing between them is the whole art.

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1. Let $L = D^2 + aD + b$ have real characteristic roots $\lambda_1 \leq \lambda_2$. Prove that $Lf \geq 0$ on $[0, \infty)$ together with $f(0) = f_0$, $f'(0) = f_1$ forces $f \geq y$, where $Ly = 0$ with the same initial data, by factoring L into two first-order operators. Deduce the estimate $f \geq 3e^{2x} - 2e^{3x}$ for $f'' - 5f' + 6f \geq 0$, $f(0) = 1$, $f'(0) = 0$ — that special case was solved directly in the homework of Differentiable functions – 1 — and show that the statement fails when the roots are complex.
 2. (Kapacëв 4.3) Let f be continuously differentiable on (a, b) with $\lim_{x \rightarrow a^+} f(x) = +\infty$, $\lim_{x \rightarrow b^-} f(x) = -\infty$ and $f'(x) + f^2(x) \geq -1$ for all $x \in (a, b)$. Prove that $b - a \geq \pi$.
 3. Prove the maximum principle for $y'' + qy \geq 0$ with $q \leq 0$, and deduce that the boundary value problem $y'' = q(x)y$, $y(a) = y(b) = 0$ with $q \geq 0$ has only the zero solution.
 4. Prove the second-order comparison theorem: if $y'' \geq F(x, y)$, $z'' = F(x, z)$ with F non-decreasing in y , and $y(a) \geq z(a)$, $y'(a) \geq z'(a)$, then $y \geq z$ on the whole interval of existence; show by an example that monotonicity in y cannot be dropped.
 5. Let y be a non-trivial solution of $y'' + q(x)y = 0$ with two zeros in $[a, b]$. Prove Lyapunov's inequality $\int_a^b q^+(x) dx > \frac{4}{b-a}$, and deduce a lower bound for the distance between consecutive zeros in terms of $\max q$.